

Transition: DS + Algorithms

Induction
& Recursion



Last time:

$$A - (B \cup C) = (A - B) \cap (A - C)$$

Take $x \in A - (B \cup C)$

by defn of $-$, $\Rightarrow x \in A$ and $x \notin B \cup C$

 $\Rightarrow x \in A$ and $[x \notin B \text{ and } x \notin C]$

All ands! Rearrange:

$$x \in A \text{ and } x \notin B \Rightarrow x \in A - B$$

$$x \in A \text{ and } x \notin C \Rightarrow x \in A - C$$

$$\text{so: } x \in A - C \text{ and } x \in A - B$$

$$\Rightarrow x \in (A - C) \cap (A - B)$$

Back to induction:
Worksheet?

Induction is not just for numbers!

Prove:

If S is a finite set with k elements, then $P(S)$ has size 2^k .

Cont

Another

Suppose n friends have a water balloon fight. Each moves to some location (all distinct), and then throws the balloon at their nearest person.

Claim:

If n is odd, then at least 1 person stays dry.

Cont:

Gossip Problem:

- n total people, each know a secret
- Every time 2 people call each other, they share every secret they know

How many phone calls are needed for everyone to know all n secrets?

Try some small ones:

Claim: If $n \geq 4$, then $2n-4$ suffice.

proof: induction on $n = \#$ of people

base case: 4 people.

Ind hyp: for $n-1$ people, can solve with $2(n-1)-4$ calls

Ind step:

Next time: Recursion

Example:

Perfect binary trees can be defined recursively:

Basis: Single vertex, height = 0

Recursive rule: given any perfect binary tree of height $k \geq 0$, build one of height $k+1$:

Theorem:

Every perfect binary tree has $2^k - 1$ vertices for some positive $k \in \mathbb{Z}$.